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(1) Research Program/Project/Study Title:

**PROFILING AND CHARACTERIZATION OF RABBITS (*Oryctolagus cuniculus*)
IN CAPIZ PROVINCE**

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Program or Project has component (if no proceed to 2)? ☐ Yes ☐ No

Component Title:	Study 1: Socio-Demographic Profile of Rabbit Growers in Capiz Study 2. Phenotypic and Morphometric Characterization of Rabbits
Component Leader/s:	N/A
Academic Rank:	N/A

(2) Category (please check)

- ☒ Social Science/Education /Humanities & other related fields
☒ Natural/Biological Sciences
☐ Food Research
☐ Developmental/ Computer/Engineering/ Architecture
☐ Extension

(3) Research Center

[Check the concerned center]

☐ SSITRDC
 Social Science and
 Industrial Technology
 Research Development
 Center

☒ LRDC
 Livestock Research
 Development Center

☐ FFRDC
 Fisheries and Food
 Research Development
 Center

☐ CSRDC
 Crop Science Research
 Development Center

(4) Site of implementation

Municipality/City:	
Province:	Capiz

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Region:		Region VI
(5) Classification Research: Development: ☒ Basic ☐ Pilot Testing ☐ Applied ☐ Tech. Promotion/Commercialization		(6) Mode of Implementation ☐ Single Authorship/Proponent ☒ Collaborative Research (within campus) ☒ Collaborative Research (intercampus) ☐ Collaborative Research (external agency)
(7) Sector/Commodity Social Rabbits Agriculture		(8) Animal Science

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(9) Rationale (State the research problem and significance of the project to the current needs of the country. The proposal should justify resource expenditure. A typical justification would include a brief introduction, a general statement concerning the historical basis for R & D and, thrusts being addressed, utilization of the expected output and the impact information generated will have on science, the target users and the country.)

Rabbits can be considered as one of the several species quite suitable for meat production (Akinsola *et al.*, 2014). They are characterized by quality protein (20 - 21%) having high biological value amino acids, meat low in calories and fat, while the animal is highly prolific with short gestation length, initial capital outlay is minimal, great genetic flexibility, lipids are highly unsaturated (60% of the total fatty acids), valuable animal model for biomedical research, low sodium and cholesterol level and a delicacy for health food products (Dalle, 2002).

Breed characterization of livestock is the first approach to a sustainable use of its animal genetic resource (Lanari *et al.*, 2003). Presently, the Philippine national livestock census barely contains breed-level data and recurring reports on breed and population statistics of rabbits as well as identified farms that engage on rabbit farming in the country. Furthermore, there is no existing record of breed level data of rabbits in the Philippines that is currently recorded in the Domestic Animal Diversity Information System (DAD-IS) of Food and Agriculture Organization of the United Nations. Anent to the abovementioned non-existence of essential data, it is imperative that it is beyond necessary for local and regional coordinating institutions and organizations as well as supporting national policies to facilitate and implement actions on animal genetic resources pursuits on breed identification and census specifically on rabbits.

Phenotypic characterization of animal genetic resources for food and agriculture (AnGR) is the practice of systematically documenting the observed characteristics, geographical distribution, production environments and uses of these resources. The information provided by characterization studies is essential for planning the management of AnGR at local, national, regional and global levels. The *Global Plan of Action for Animal Genetic Resources* recognizes that “A good understanding of breed characteristics is necessary to guide decision-making in livestock development and breeding programs” (CGRFFA 2011).

Characterization of animal genetic resources for food and agriculture (AnGR) involves three types of information: phenotypic, genetic and historical. The weight given to each depends on the country (e.g. whether it is developed or developing) and the objective (e.g. improvement, conservation or breed differentiation). These guidelines focus on the collection and use of phenotypic information. Characterization of AnGR encompasses all activities associated with the identification, quantitative and qualitative description, and documentation of breed populations and the natural habitats and production systems to which they are or are not adapted. The aim is to obtain better knowledge of AnGR, of their present and potential future uses for food and agriculture in defined environments, and their current state as distinct breed populations (FAO, 1984; Rege, 1992). The information provided through the characterization process enables a range of interest groups, including farmers, national governments

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and regional as well as global bodies to make informed decisions on priorities for the management of AnGR (FAO, 1992; FAO/ UNEP, 1998).

The term “phenotypic characterization of AnGR” generally refers to the process of identifying distinct breed populations and describing their external and production characteristics within a given production environment. In these guidelines, the definition is broadened to include the description of the production environment. It is used to identify and document diversity within and between distinct breeds, based on their observable attributes. The term “production environment” is here taken to include not only the “natural” environment but also management practices and the uses to which the animals are put, as well as social and economic factors such as market orientation, niche-marketing opportunities and gender issues. Recording the geographical distribution of breed populations is here considered to be an integral part of phenotypic characterization. Complementary procedures used to unravel the genetic basis of phenotypes and their patterns of inheritance from one generation to the next, and to establish relationships between breeds, are referred to as molecular genetic characterization (FAO, 2011b).

As a step towards the inclusion of breed-level data in national livestock censuses, the next World Programme for Census of Agriculture (produced by FAO every ten years to guide countries in conducting of their agricultural census) (FAO, 2006) encourages countries to collect and report livestock data at breed level. Within the framework of the CBD, countries are obliged to report on the status of their national biodiversity. It has also been agreed that countries should report on their progress in the implementation of the *Global Plan of Action for Animal Genetic Resources* every four years by submitting country reports to FAO (FAO, 2009a).

Without good knowledge of AnGR, decision-makers at national level and in the breeding industry will be unable to develop strategic plans for sustainable use and development. Knowledge of all the breeds that might be drawn upon, and of the production environments in which the animals can be kept, is needed in order to develop or strengthen animal breeding programs (FAO, 2010a). The information gained from monitoring through repeated surveys is important for identifying trends that need to be addressed in future plans, and provides a basis for assessing progress in the implementation of existing plans and policies.

To date, there is no well-integrated information system of rabbit breeds and population in the province to account them and only a few of local and national actions in animal resources management have been conducted and reported in rabbits. Moreover, established baseline and specific guidelines for rabbit characterization is fragmented and has not been established to date. Hence, this study is proposed to justify the set objectives and establish a baseline on rabbit characterization in the local level and provide additional data for future studies that will serve as a reference for wider scope of rabbit genetic resource management in the national and international level for future researches. Additionally, this study also aims to aid in developing practical and effective production management

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practices that could potentially improve rabbit production in different schemes in the local level and contribute to understanding of the farmers' attitude and preference towards rabbit meat consumption.

(9-A) Project Description (Discuss the project strategy and plan of action of the project. It should also describe the technical procedures and other works needed to carry out the project objectives. The relationship of the project inputs and activities should be explained clearly in this section. Projects similar/related to the proposed project should be mentioned.)

(10) Objectives/Statement of the Problem (for Social Sciences) (Statement of the specific purpose to address the problem areas of the project.)

STUDY 1: SOCIO-DEMOGRAPHIC PROFILE OF RABBIT GROWERS IN CAPIZ

The study will focus on the socio-demographic profile rabbit growers and value chain analysis of rabbit growers in Capiz province and specifically aims to address and obtain responses on the following questions:

1. What are the socio - demographic aspects of rabbit growers and raisers in the province of Capiz in terms of gender, age bracket, educational attainment, employment status and period of raising rabbits.
2. What are the different adapted practices of rabbit raisers in terms of housing, breeding, feeding management, confinement and rearing systems, type of production system, marketing, pricing and identified niche markets
3. What are the personal purpose, perception, and attitude of rabbit growers towards rabbit meat in terms of consumption?
4. What are the specific support activities and interventions being provided by the government to rabbit raisers in Capiz.
5. What are the common problems, production constraints encountered by the rabbit growers and raisers in the province and recommendations to address such constraints.
6. What is the geographic diversity of rabbit farms and growers in the province of Capiz

STUDY 2: PHENOTYPIC CHARACTERIZATION OF RABBITS IN CAPIZ PROVINCE

Objectives:

1. Obtain population record and identify existing rabbit breeds that commonly raised by rabbit raisers and growers in the province.
2. Measure the phenotypic characteristics of rabbit populations in Capiz through evaluation of external body measurements, physical features, body weight and dimensions.

(11) Scientific Basis/Theoretical Framework (Include concepts, theories and variables that indicate relevant relationships to be tested.)

STUDY 1: SOCIO-DEMOGRAPHIC PROFILE OF RABBIT GROWERS IN CAPIZ

The study will be anchored on Michael Porter's Value Chain Model (1985) where he indicated that a value chain is a collection of activities that are performed by a company to create value for its customers. Moreover, Chai (2022) states that the "value chain is a concept describing the full chain of

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a business's activities in the creation of a product or service – from the initial reception of materials all through its delivery to market and everything in between”.

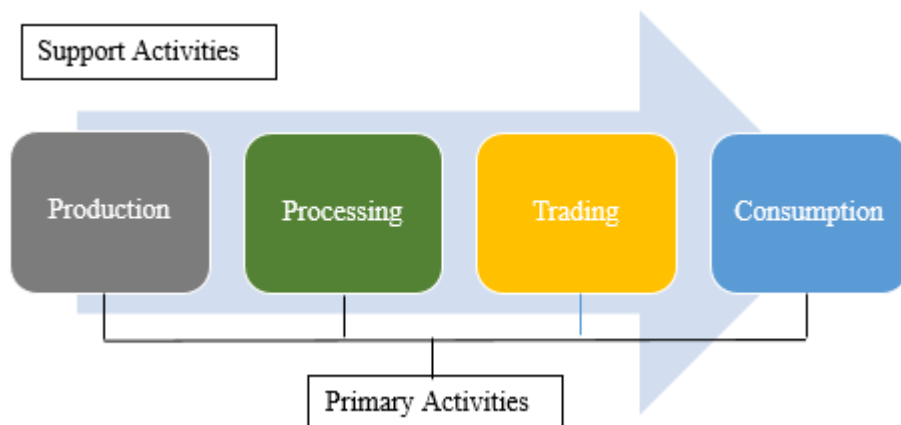


Figure 1. Theoretical and Conceptual Framework of the Study

(12) Review of Literature

Breed characterization of livestock is the first approach to a sustainable use of its animal genetic resource (Lanari *et al.*, 2003). The first knowledge of the characterization of local genetic resources on the knowledge of morphological traits (Delgado *et al* 2001). Morphometric measurements have been found useful in contrasting size and shape of animals (Ajayi *et al.*, 2008). Growth, which is the totality of the bodily development that can be seen phenotypically is the main economic trait needed for improved profitability for farmers and adequate meat for the consumers.

Characterization of quantitative traits has been performed on Hycole, Hyla rabbits for litter size, growth rate, and morphometrics of the body (Setiaji *et al.*, 2012). Crosses Hycole rabbit with NZW and their reciprocal has been done with the crossed Hycole sire and NZW dam (P_{AN}) (Brahmantiyo *et al.*, 2017). The rabbit has potential to be developed as the genetic resources of dam adaptive to tropical climate due to its good heterosity (hybrid vigor) in litter size and growth rate (Khan *et al.*, 2017a, 2017b; Tariq *et al.*, 2015).

A survey of the native rabbit populations was conducted in three governorates, FY, BN, and MN in a period from December 2017 to February 2018 (Figure-1). The phenotype data and blood samples of Native Middle Egypt rabbits (NMERs) were collected randomly from six localities of the three governorates. The phenotypic characterization of NMER was made according to Khalil (2010) and included the profile body of the head, ears, eyes, neck, and legs and the coat colors. The morphological characteristics of NMER populations were taken by body weight and body dimension measurements on 83 mature rabbits. The body length was measured from atlas to the first coccygeal vertebrae, while the chest circumference was measured behind the shoulder blades and the abdominal circumference at the level of the seventh lumbar vertebrae. The ear width was from the outside to another side of the ear. Ear length was taken from the bottom to the top of the ear, and foot length was from behind the foot to start the fingers.

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Phenotypic and genetic characterizations of the native rabbits are essential studies for the identification and improvement of the breeding programs and to help for their conservation (Galal and Khalil 1994). The population genetic structure is the result of many factors such as geographical and ecological factors which may cause variation and division of the populations. Moreover, the genetic drift, gene flow, and the balance between them can also result in population genetic structure (Abdel-Kafy et al (2016); (Oseni and Oke (2012)).

An integrated phenotypic and genetic characterization study of three meat-type goat breeds developed in South Africa (Boer, Savanna and Kalahari Red) was conducted in 2007 to determine whether the typical characteristics of the breeds were being maintained and ensure that their unique traits were not being lost. The same populations were sampled for phenotypic and genetic characterization. A set of twelve linear measurements were taken for the phenotypic characterization. Eighteen microsatellite markers selected from a panel of markers recommended by the International Society for Animal Genetics (ISAG) and FAO were used in the genetic characterization. The results showed that morphometric variation within the breeds is greater than between the breeds, and that morphometric differences between the breeds were fairly insignificant. This highlighted the need for genetic characterization to enable the breeds to be distinguished accurately at genotypic level. Results of the genetic studies showed that the three breeds had relatively high heterozygosity values and that each of the populations was clearly distinguishable as a separate breed based on genotyping results with the selected markers. It was concluded that further genetic studies were needed to ensure sufficient diversity within the breeds for long-term conservation of the unique genetic resources.

(13) Methodology (Details such as; a) variables to be measured and evaluated; b) treatments to be used and their layout; c) experimental procedures and design; d) statistical analysis; e) evaluation method and observations to be made, strategies for implementation.)

STUDY 1: SOCIO-DEMOGRAPHIC PROFILE OF RABBIT GROWERS IN CAPIZ

Identification of Rabbit Farms and Growers in the Province

A letter from the Satellite College Director requesting to visit the Department of Agriculture office of the Province of Capiz will be secured to gather secondary data and information about the status of rabbit production in the whole province. All data and information to be obtained from the office will be purposely utilized only for research grounds. After successful gathering of relative and relevant information needed from the office, the selected areas for conducting the research will be identified from the municipalities of the Capiz down to its respective barangays with the most number of existing rabbit growers and farms.

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Population, Sampling Size and Sampling Technique

The descriptive method of research will be used because; it describes and interprets what is. It concerns with conditions of relationships that exist; practices that prevail; beliefs, processes that are going on; effects that are being felt, or trends that are developing (Calderon and Gonzales, 1993). A survey of rabbit populations will be conducted in five municipalities in the province with the most number of rabbit growers and raisers. Succeeding five barangays in each municipality will be identified and ten farms/raisers every barangay will serve as respondents. Purposive sampling method will be used. The non-scientific or non-probability sampling method through purposive sampling will be used.

Location of the Study

The study will be conducted in the province of Capiz covering the five municipalities with the most number of rabbit farms and growers down to the identified barangays of each respective municipality.

Method of Data Collection

Data collection will be done through questionnaires to serve as an instrument in the data-gathering process and instruments will be validated by social science experts of the campus. The instruments will be distributed to the respondents to answer the questions followed by unstructured interview. Interview will be constructed in English and translated into the local dialect in order to clearly deliver and clarify to the respondents the contained statements of the questionnaires to the respondents. Data collection will commence as soon as necessary permit and clearances from LGU offices and officials is secured. To facilitate the conduct of data gathering on the specific municipalities and barangays, personal consultation with the officials for data collection will also be done for assistance during the conduct of the field interview. The researchers and research assistants will serve as the data accumulators. Retrieval of questionnaire will be done immediately to minimize the loss of data.

Analysis of Data

After all the interviews will be completed, the data and information will be extracted from the instruments which will then be encoded for statistical analyses. Since this study is purely descriptive in nature, purposive sampling as non-probability sampling method will be used and the data to be accumulated will be summarized and analyzed using descriptive statistics such as frequencies, percentages, means, and standard deviation.

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STUDY 2: PHENOTYPIC CHARACTERIZATION OF RABBITS IN CAPIZ PROVINCE

Objectives:

1. Identify and record the existing rabbit population and breeds that are commonly raised by the growers in the province of Capiz.
2. Identify and document the phenotypic characteristics of rabbits through assessment of their respective physical features specifically; body type, coat color, ear type and eye color.
3. Measure the morphological characteristics of rabbit populations in Capiz by obtaining the body weight and body dimension measurements.

Data Gathering Procedure

The phenotypic characterization of rabbits will be done according to (Khalil 2010) and will include the profile body of the head, ears, eyes, neck, and legs and the coat colors. The morphological characteristics of rabbit populations will be obtained by body weight and body dimension measurements. The traits will be collected based on visual observation and actual measurements. High resolution digital camera will be used to capture images of animal as well as subjective qualitative data (e.g. body shape, ear type, eye and coat/fur color). Tape measure will be used to obtain linear body dimension measurements. The body length will be measured from atlas to the first coccygeal vertebrae, while the chest circumference will be measured behind the shoulder blades and the abdominal circumference at the level of the seventh lumbar vertebrae. The ear width is from the outside to another side of the ear. Ear length will be taken from the bottom to the top of the ear, and foot length is from behind the foot to start the fingers. The data to be gathered will be recorded using a data sheet.

Sampling Technique

Qualitative and quantitative data on phenotypic and morphometric characteristics of rabbits will be obtained. The non-scientific or non-probability sampling method will be used. This will be employed because in non-probability sampling, not all of the individuals in a population are given equal chance of being included in the sample. Hence, the sample will be selected based on subjectivity. A total of two hundred mature doe and buck will be subjected to morphometric evaluation by taking body weight and body dimension measurements.

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Phenotypic Traits	Morphometric Traits
Body type	Body weight
Coat Color/Fur	Body length
Eye Color	Chest circumference
Ear type	Ear length
	Ear width
	Height

Table 1. Metrics for Phenotypic and Morphometric Traits

Analysis of Data

The quantitative data to be gathered will be statistically analyzed using the Special Program for Social Sciences (SPSS) version 2.0. The statistical tools to be employed in analyzing the data will be as follows: frequency count/percentages and means are the statistical tools to be used to identify the phenotypic and morphometric characteristics of rabbits.

(14) Major Activities/Work plan (Enumerate in chronological order the task to be undertaken. Use gantt chart. Below is an example. This can be modified base on your desired Work plan.)

Research Objectives	Year 1										Year 2	
	Nov	Dec	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug		
1. Preparation and presentation of research proposal												
2. Official visit to DA provincial office for inquiries and gathering of preliminary substantial data down local LGU's and barangays for assistance.												
3. Preparation of checklist/questionnaires for intended data to be accumulated												
4. Hiring of data entry staff/enumerators, research assistant.												

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SOURCES OF EXPENSES	QUANTITY	UNIT	PRICE PER UNIT	TOTAL
SUPPLIES				
Printer (Epson L3150)	1	pc	9,500	9,500
Digital camera (Canon EOS 600D)	1	pc	22,000	22,000
Digital Weighing scale	1	pc	3,000	3,000
A4 size bond paper	10	reams	350	3,500
Black and colored inks	6	bottles	300	1,800
Tape measure	5	pcs	50	250
Recording Notebook	1	pc	80	80
Ballpen	10	pcs	30	300
SUBTOTAL				40,430.00
Personnel Services (For studies needed the Research Assistant, informant/s, guide)				
Research Assistant/	Days	120	500	60,000.00
Travel Allowance	Days	120	150	18,000
SUBTOTAL				78,000.00
TOTAL EXPENSES				118,430.00

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(20) Capsule Curriculum Vitae (One-page brief C.V. only)

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